

SECTION A

- 1 B. Enabled a new process and improved information flow IT allows doctors, nurses, and pharmacists to access records simultaneously, improving information flow and enabling new business processes.
- 2 B. E-Commerce E-commerce refers to buying and selling goods and services over the Internet.
- 3 C. Outsourcing The university hires an external software company instead of developing the system internally.
- 4 C. Off-the-Shelf Software Purchasing already available commercial software is known as buying off-the-shelf software.
- 5 C. Technology Servers, software, databases, and networks are all part of the technology dimension of an Information System.
- 6 B. Organizational Dimension Changing departments and reporting structures relates to the organizational structure of the business.
- 7 C. ESS (Executive Support System) ESS provides executives with summarized information, dashboards, market trends, and strategic reports.

Final Answers

1. B
2. B
3. C
4. C
5. C
6. B
7. C
8. B

SECTION B

Question 1

(a) Briefly explain the six (6) strategic business objectives for investing in Information Systems.

1. Operational Excellence

Improves efficiency and productivity while reducing operating costs.

2. New Products, Services and Business Models

Enables organizations to create innovative products, services, and new ways of doing business.

3. Customer and Supplier Intimacy

Improves relationships with customers and suppliers, increasing satisfaction and reducing costs.

4. Improved Decision Making

Provides accurate and timely information to support better managerial decisions.

5. Competitive Advantage

Helps the organization perform better than competitors through lower costs, better services, or innovation.

6. Survival

Enables organizations to meet legal requirements and remain competitive in the industry.

(b) Discuss three reasons how Information Systems provide competitive advantage.

1. Increase operational efficiency, reducing costs and improving productivity.
2. Improve customer service by providing faster, more personalized services, increasing customer loyalty.
3. Support better decision-making through real-time and accurate information, allowing faster responses to market changes.

Question 2

(a) Describe the three dimensions of an Information System.

1. Organizational Dimension

Includes people, organizational structure, culture, business processes, and policies.

2. Management Dimension

Involves managers who plan, organize, make decisions, solve problems, and develop strategies.

3. Technology Dimension

Includes hardware, software, databases, networks, and telecommunications used to process and communicate information.

(b) Explain how each dimension contributes to successful implementation.

Organizational Dimension

Ensures employees and business processes are aligned with the new system.

Management Dimension

Provides leadership, planning, decision-making, and support for successful implementation.

Technology Dimension

Provides reliable hardware, software, databases, and networks needed for efficient system operation.

Question 3

(a) Outline two major methods of acquiring Information Systems.

1. Buying an off-the-shelf (generic) application.
2. Building a custom (bespoke) application.

(b) Differentiate the two methods.

Off-the-Shelf Software Custom Application

Already developed software Developed specifically for one organization

Faster to implement Takes longer to develop

Less expensive More expensive

Limited customization Fully customized to business needs

(c) Discuss three advantages and three disadvantages of outsourcing information system development.

Advantages

1. Reduces development costs.
2. Provides access to experienced specialists.

3. Allows the organization to focus on its core business.

Disadvantages

1. Loss of organizational competencies.

2. Possible reduction in service quality.

3. Unexpected costs may increase project expenses.

Question 4

(a) Describe the following systems.

i. Transaction Processing System (TPS)

A system that records and processes routine daily business transactions such as sales, payroll, and inventory.

ii. Management Information System (MIS)

A system that provides summarized reports from TPS data to support routine managerial decisions.

iii. Decision Support System (DSS)

A system that supports managers in making non-routine decisions by analyzing data and models.

iv. Executive Support System (ESS)

A system that provides senior executives with summarized internal and external information for strategic decision-making.

(b) Explain how these systems are related.

TPS collects and stores daily transaction data.

MIS uses TPS data to generate reports for middle managers.

DSS uses information from TPS and MIS (and sometimes external data) to support complex decisions.

ESS receives summarized information from TPS, MIS, and DSS to help senior executives make strategic decisions.

Relationship flow:

TPS → MIS → DSS → ESS

or

TPS provides data to MIS and DSS, while ESS receives summarized information from all lower-level systems.

SECTION A

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SECTION B

5(a) Discuss five factors managers should consider when evaluating IT infrastructure. (5 Marks)

Managers should consider the following factors:

1. Adaptability

The infrastructure should be flexible enough to accommodate new technologies and changing business needs.

2. Scalability

The infrastructure should be able to grow or shrink according to the organization's needs without major changes.

3. Standardization

Hardware and software should follow industry standards to ensure compatibility, easier maintenance, and future upgrades.

4. Security

The infrastructure should protect data, hardware, software, and networks from unauthorized access, cyber-attacks, and data loss.

5. Financial Considerations

Managers should evaluate the cost of implementation, maintenance, expected benefits, risks, system life span, and return on investment (ROI) or Net Present Value (NPV).

5(b) Explain the five phases of disaster management. (5 Marks)

1. Disaster Prevention

Measures taken to prevent disasters from occurring or reduce their likelihood, such as installing fire protection systems and implementing cybersecurity measures.

2. Disaster Preparedness

Activities carried out before a disaster to improve readiness, including emergency planning, staff training, evacuation drills, and early warning systems.

3. Disaster Response

Immediate actions taken during or immediately after a disaster to save lives and reduce damage, such as rescue operations and emergency relief.

4. Disaster Recovery

Actions taken after the disaster to restore normal operations, including rebuilding infrastructure, restoring IT systems, and resuming business activities.

5. Disaster Mitigation

Long-term measures aimed at reducing the impact of future disasters, such as risk assessments, stronger infrastructure, and regular data backups.

5(c) Explain any five (5) elements of D'Aveni's Hypercompetition Model. (5 Marks)

Any five of the following seven elements are correct:

1. Superior Stakeholder Satisfaction

Providing greater value and satisfaction to customers, employees, shareholders, and other stakeholders.

2. Strategic Soothsaying

Identifying future opportunities and market trends before competitors through research and innovation.

3. Positioning for Speed

Preparing the organization to respond quickly to changes in the market and customer needs.

4. Positioning for Surprise

Introducing unexpected products, services, or strategies that competitors do not anticipate.

5. Shifting the Rules of Competition

Changing the basis of competition by introducing new technologies, products, or business models.

Signaling Strategic Intent – Communicating future strategic plans to influence competitors' decisions.

Simultaneous and Sequential Strategic Thrusts – Continuously launching multiple strategic actions to maintain a competitive advantage.